

SMARTX: Porewater Methane

July 2026 Samples

2026-08-28

Contents

1	Read in data files	2
2	Assess standard curves	2
3	Aanlayze the Check Standards	5
4	Analyze the Blanks	6
5	Dilution correct samples and Check Range	9
6	Calculate gas in water	10
7	Merge samples with Metadata & check all samples are present	10

##Run Information

```
cat("Run Information: Brenden D. Blakley ") #lets you know what section you're in
```

Run Information: Brenden D. Blakley

```
library(here)
```

here() starts at C:/Users/blakleybd/OneDrive - Smithsonian Institution/Documents/GitHub/GCREW

```
#set the run date & user name
```

```
sample_year <- "2026"
```

```
sample_month <- "July"
```

```
user <- "Brenden Blakley"
```

```
#identify the files you want to read in
```

```
#read in as a list to accommodate multiple runs in a month
```

```
GHG_files <- c("Raw Data/SMARTX_July_Run2_330-463_formatted.csv", "Raw Data/20260820_SMARTX_2026_May_610")
```

```
# Define the file path for QAQC log file - NO Need to change just check year
```

```
log_path <- here("Porewater QAQC Logs", "GCRew_CH4_QAQC_Log.csv")
```

```
# Define final path for data be sure to change year and month and run number
```

```
final_path <- "Processed Data/GCRew_SMARTX_Porewater_CH4_202607.csv"
```

```
#Gas in Water Calculations
```

```
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
```

```
TempC = 25 #Equilibration Temperature (Celsius)
```

```
volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrated
```

```
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrated (should be equal to volume_g)
```

```
#record any notes about the run or anything other info here:
```

```
run_notes <- ""
```

```
#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"
```

```
cat(run_notes)
```

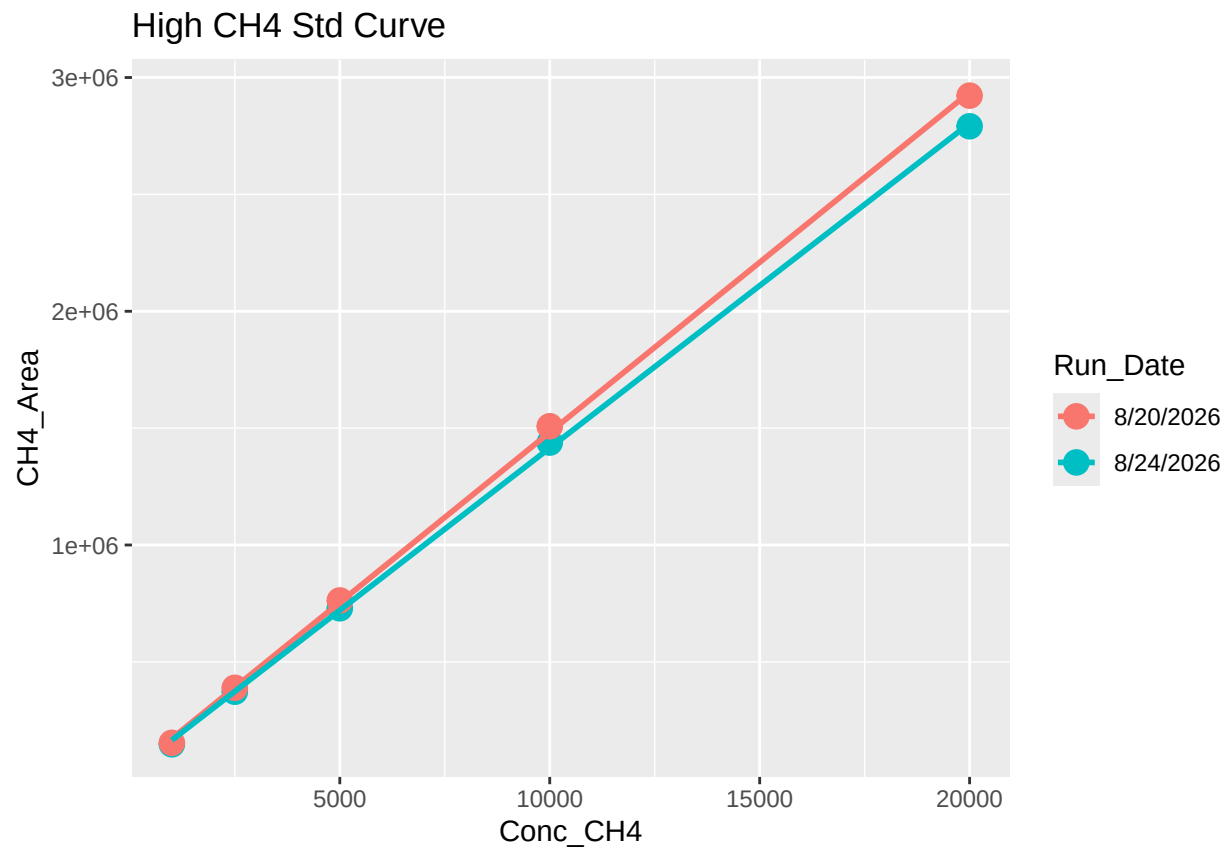
##Set Up

##Read in metadata and create similar sample IDs for matching to samples

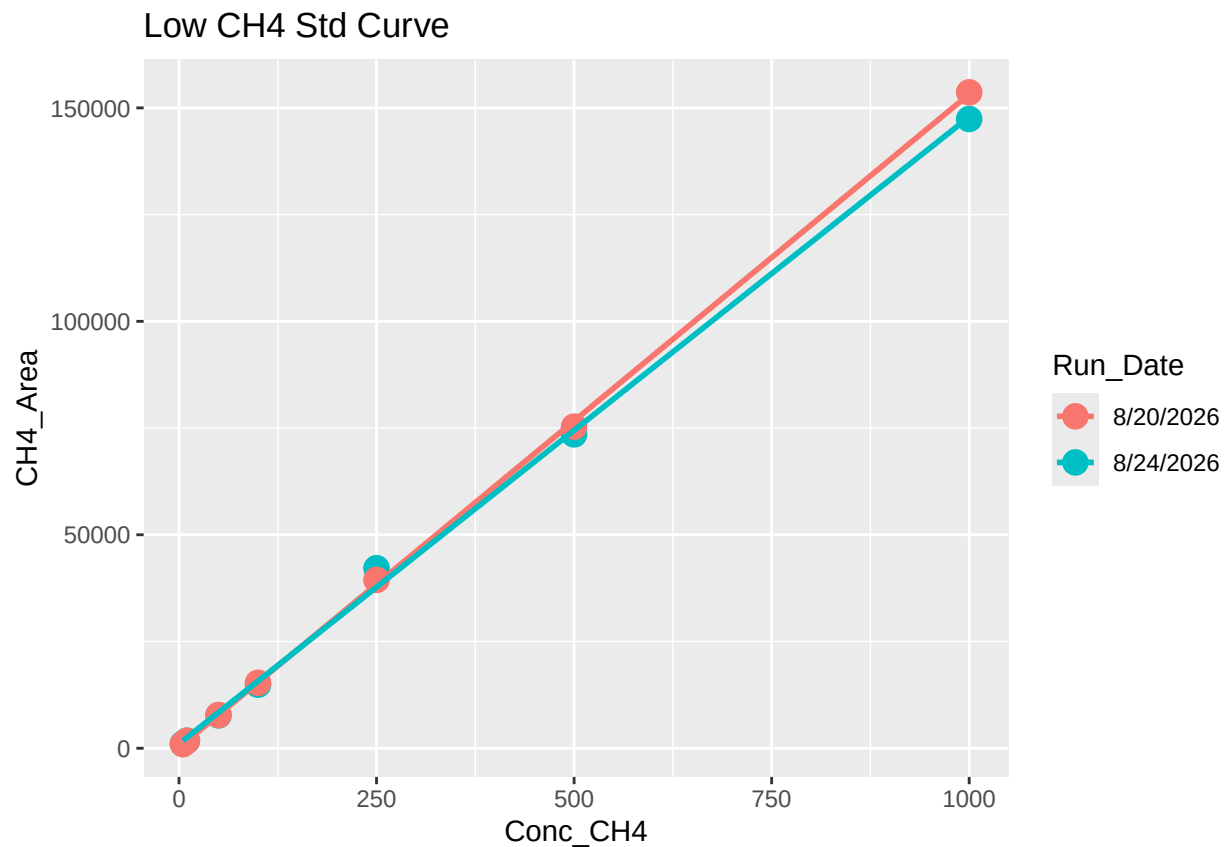
1 Read in data files

2 Assess standard curves

'geom_smooth()' using formula = 'y ~ x'

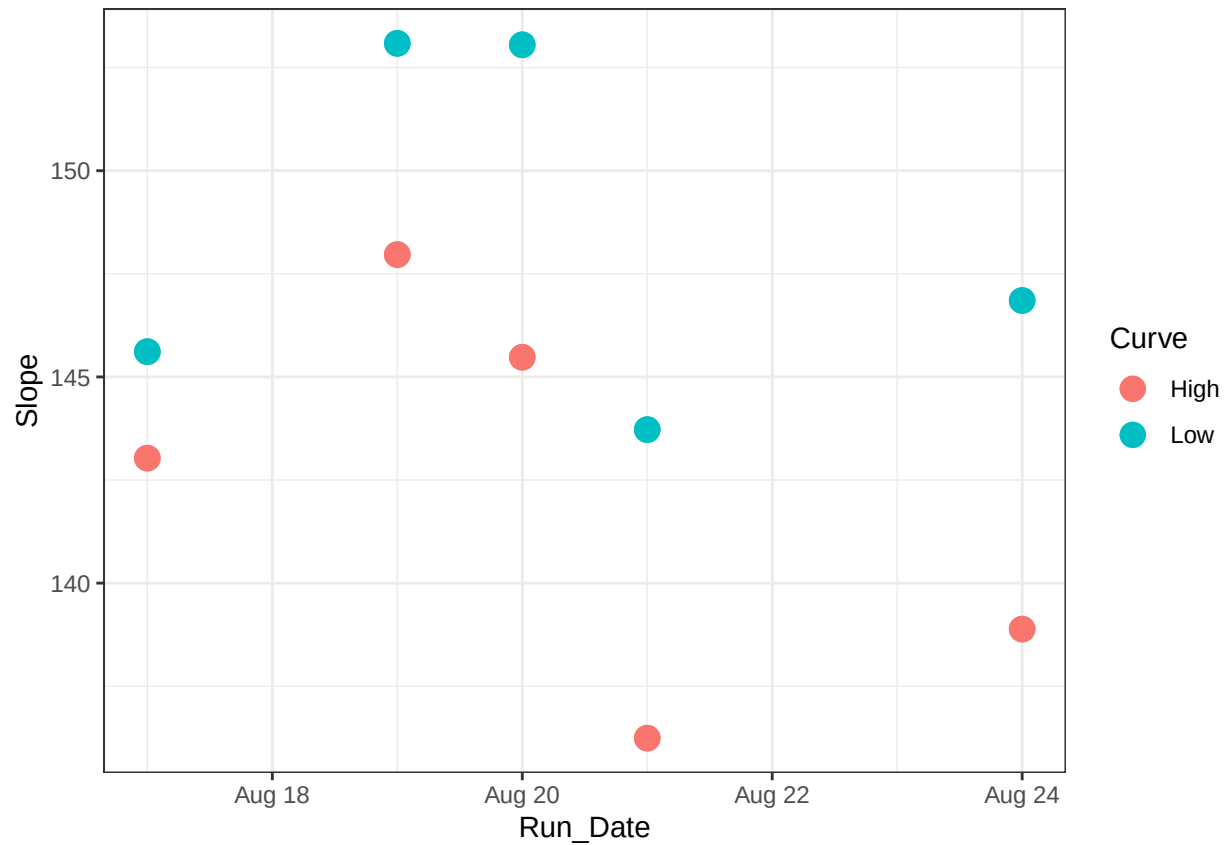


```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
##      Curve      R2      Slope Intercept Run_Date
## 1   High 0.9994458 136.2406 25980.5484 8/21/2026
## 2    Low 0.9995311 143.7214   801.9223 8/21/2026
## 3   High 0.9999168 147.9666 17873.3851 8/19/2026
## 4    Low 0.9997966 153.0888   407.2899 8/19/2026
## 5   High 0.9999646 143.0303  7587.2742 8/17/2026
## 6    Low 0.9996977 145.6125   745.1636 8/17/2026
```

```
## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'Run_Date = mdy(Run_Date)'.
## Caused by warning:
## ! 4 failed to parse.
```



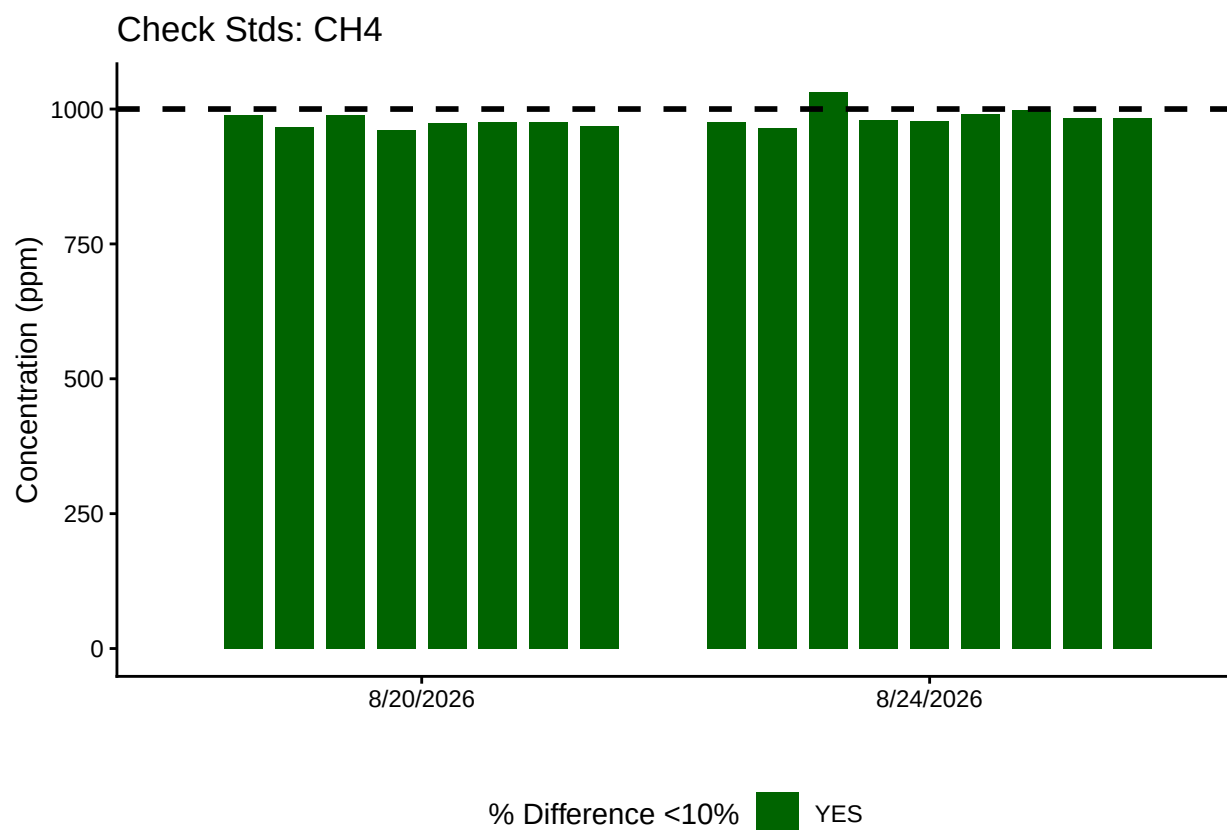
```
## [1] "High_R2 is Good-8/20/2026" "High_R2 is Good-8/24/2026"
```

```
## [1] "Low_R2 is Good-8/20/2026" "Low_R2 is Good-8/24/2026"
```

2.1 Now calculate the CH4 in ppm

3 Aanlayze the Check Standards

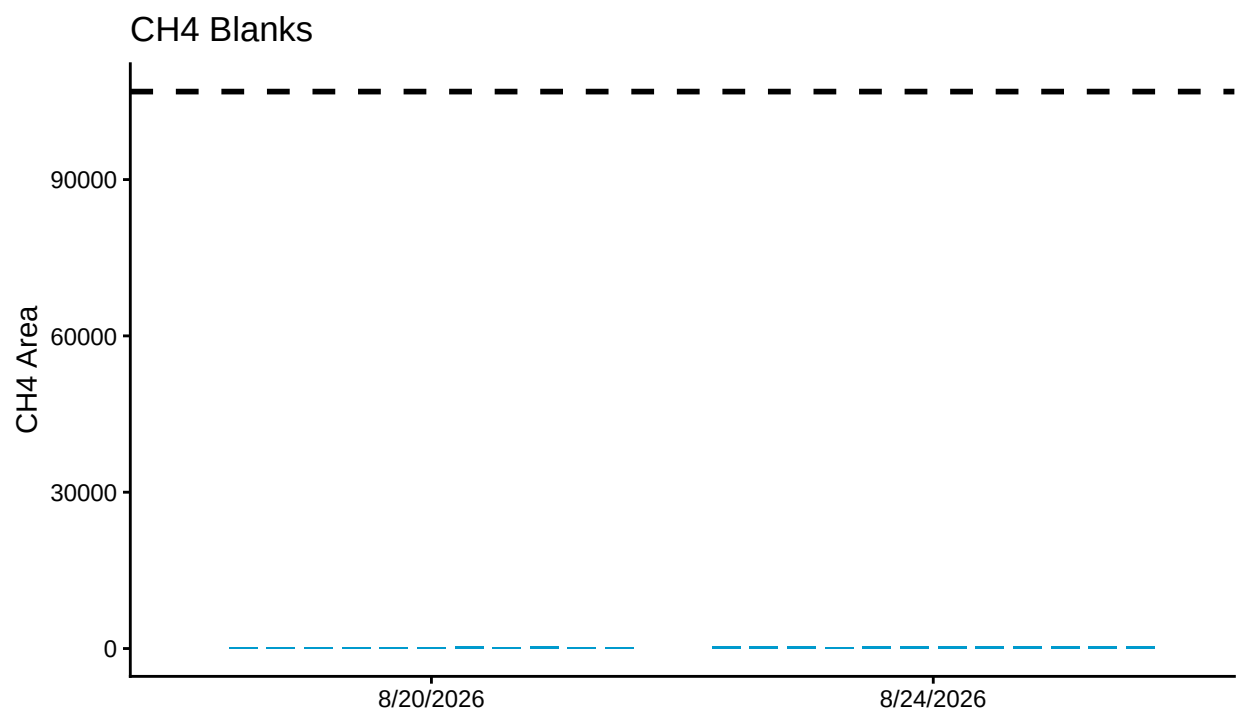
```
## >60% of CH4 Check Standards are within range of expected concentration - PROCEED
```



4 Analyze the Blanks

Assess Blanks

>60% of CH4 Blanks are below the lower 25% quartile of samples - PROCEED

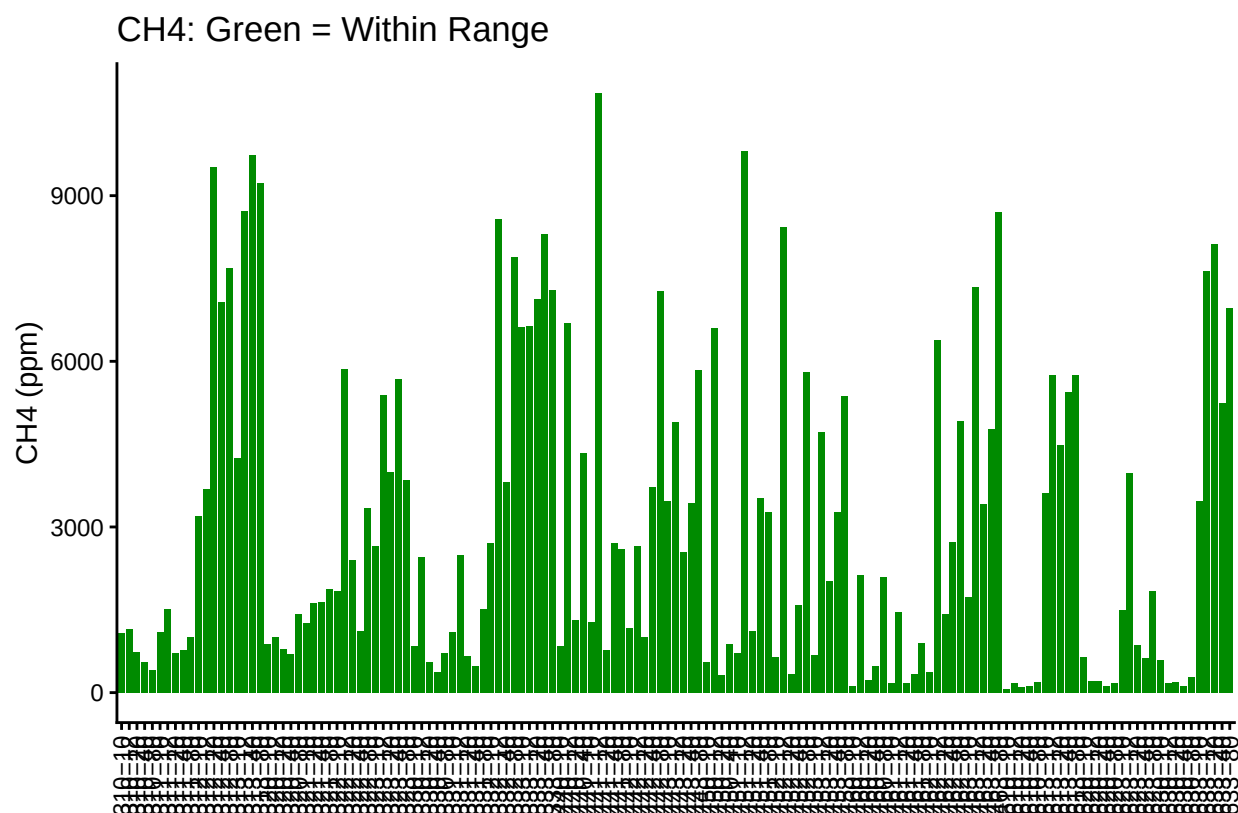
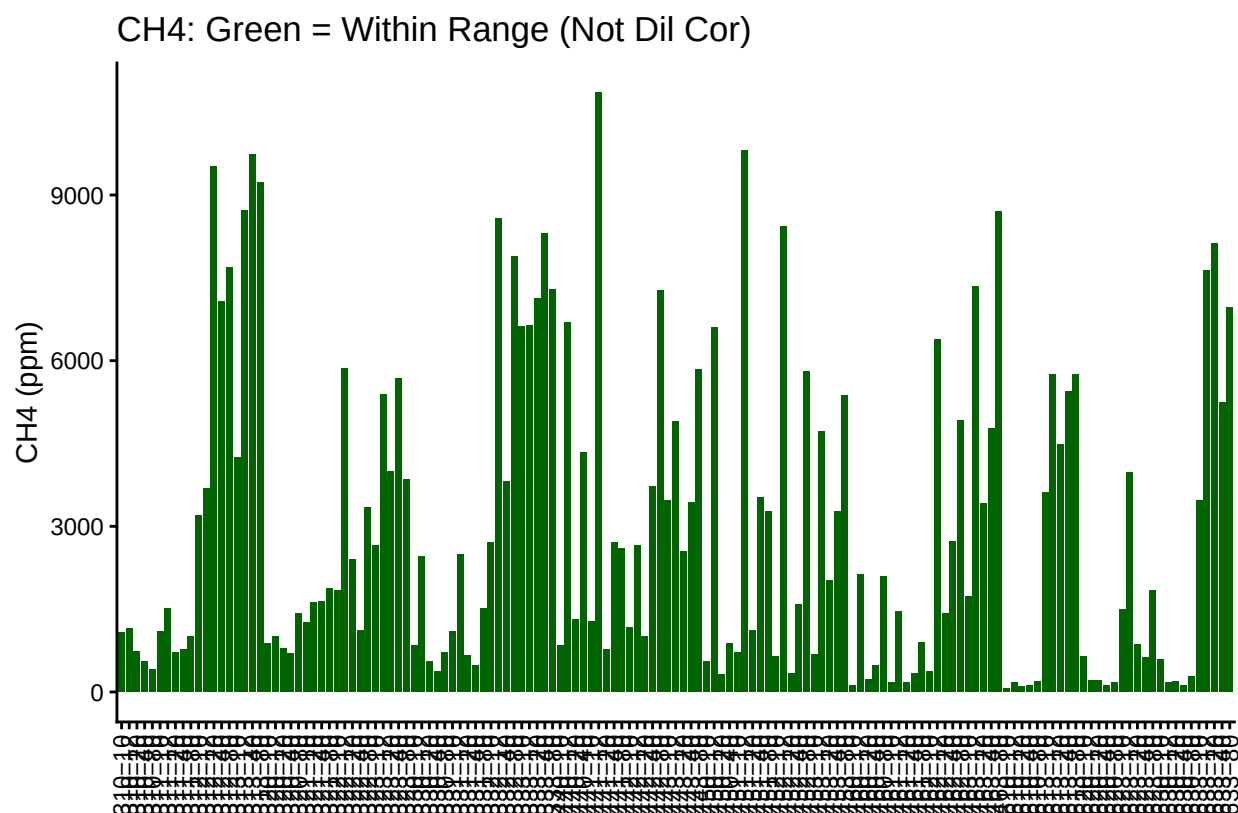


Blank Conc <25% Quartile Samples ■ YES

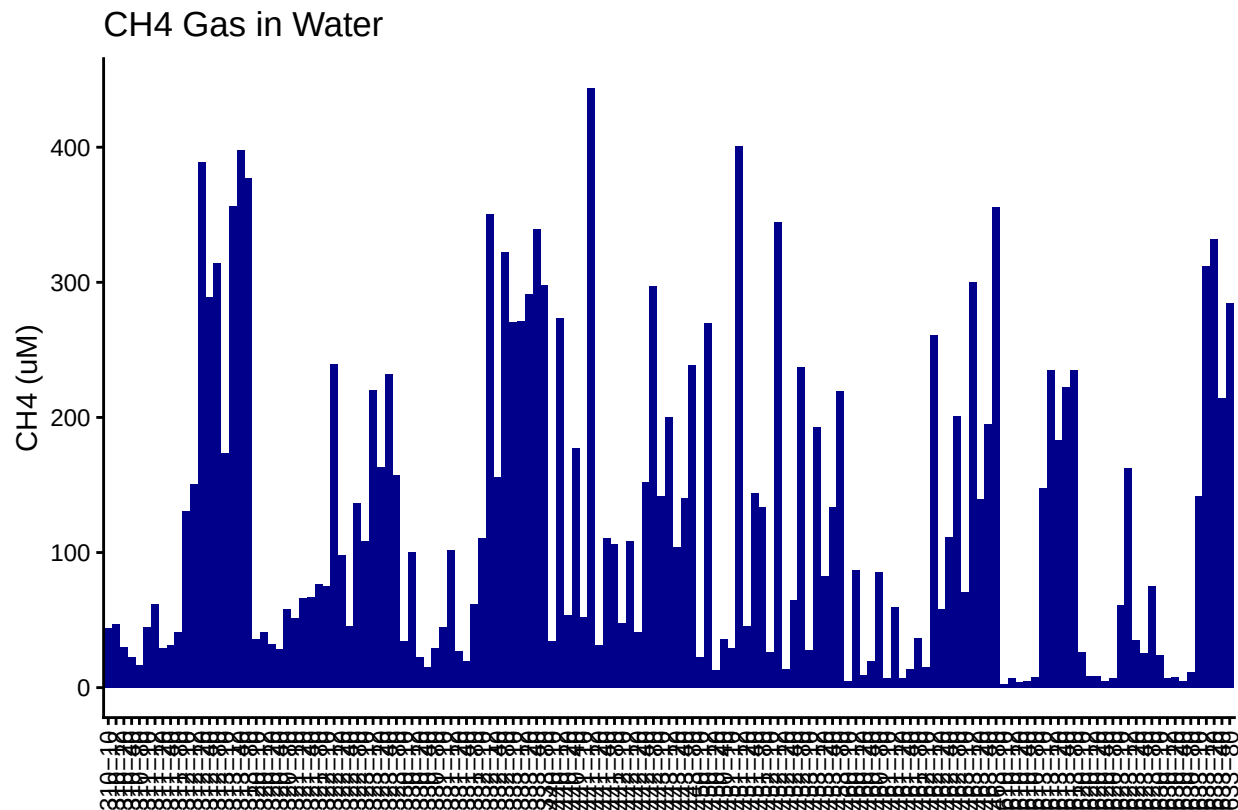
Table 1: Mean Area of Blanks

Test	Blank_Mean_Area
CH4	388.0435

5 Dilution correct samples and Check Range



6 Calculate gas in water



7 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata
```

```
## Some sample IDs are missing from data.
```

```
## [1] "SMARTX_313-20" "SMARTX_332-20" "SMARTX_440-80" "SMARTX_450-80"
```

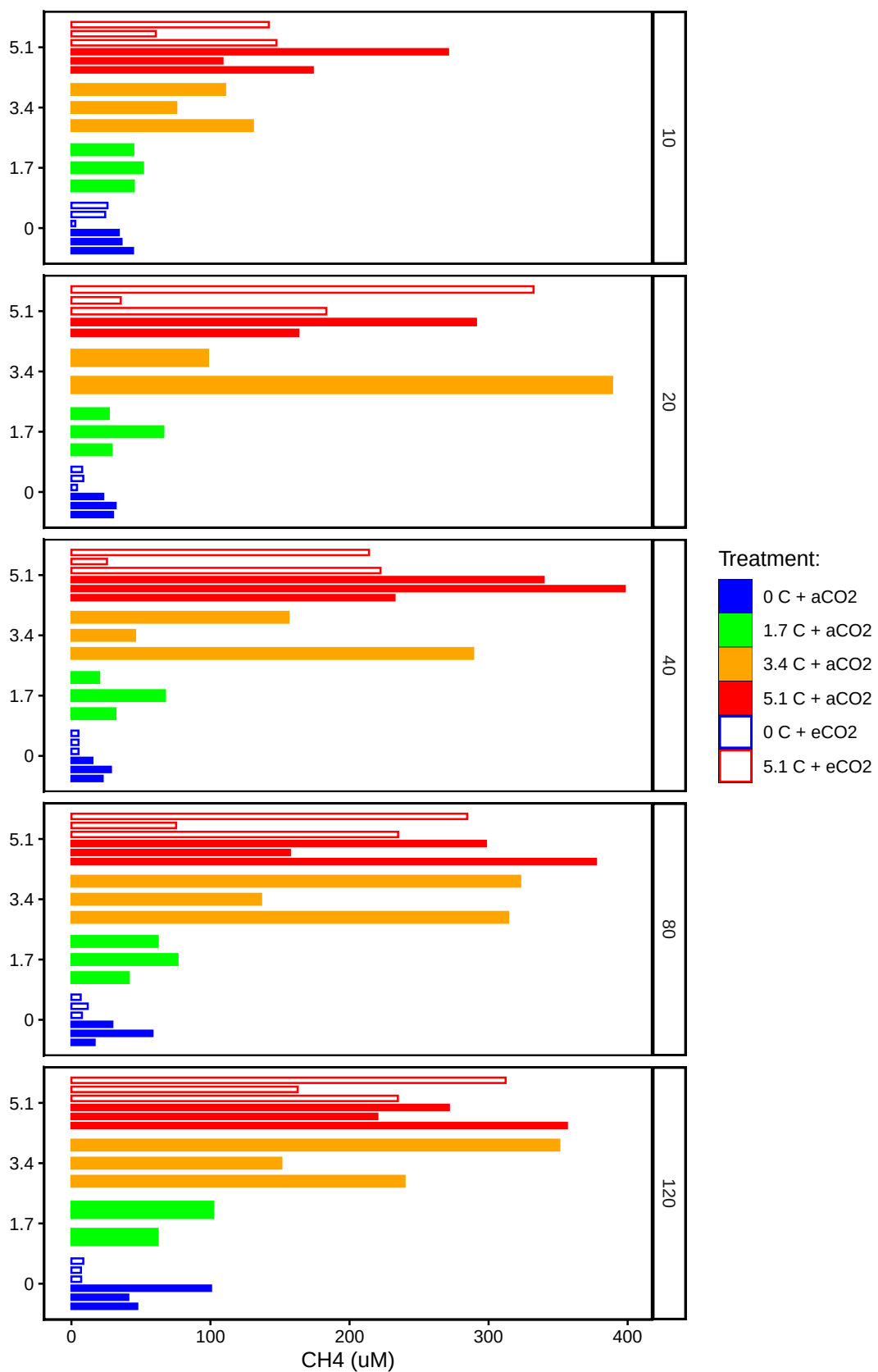
```
## [5] "SMARTX_321-120"
```

7.1 Organize Data

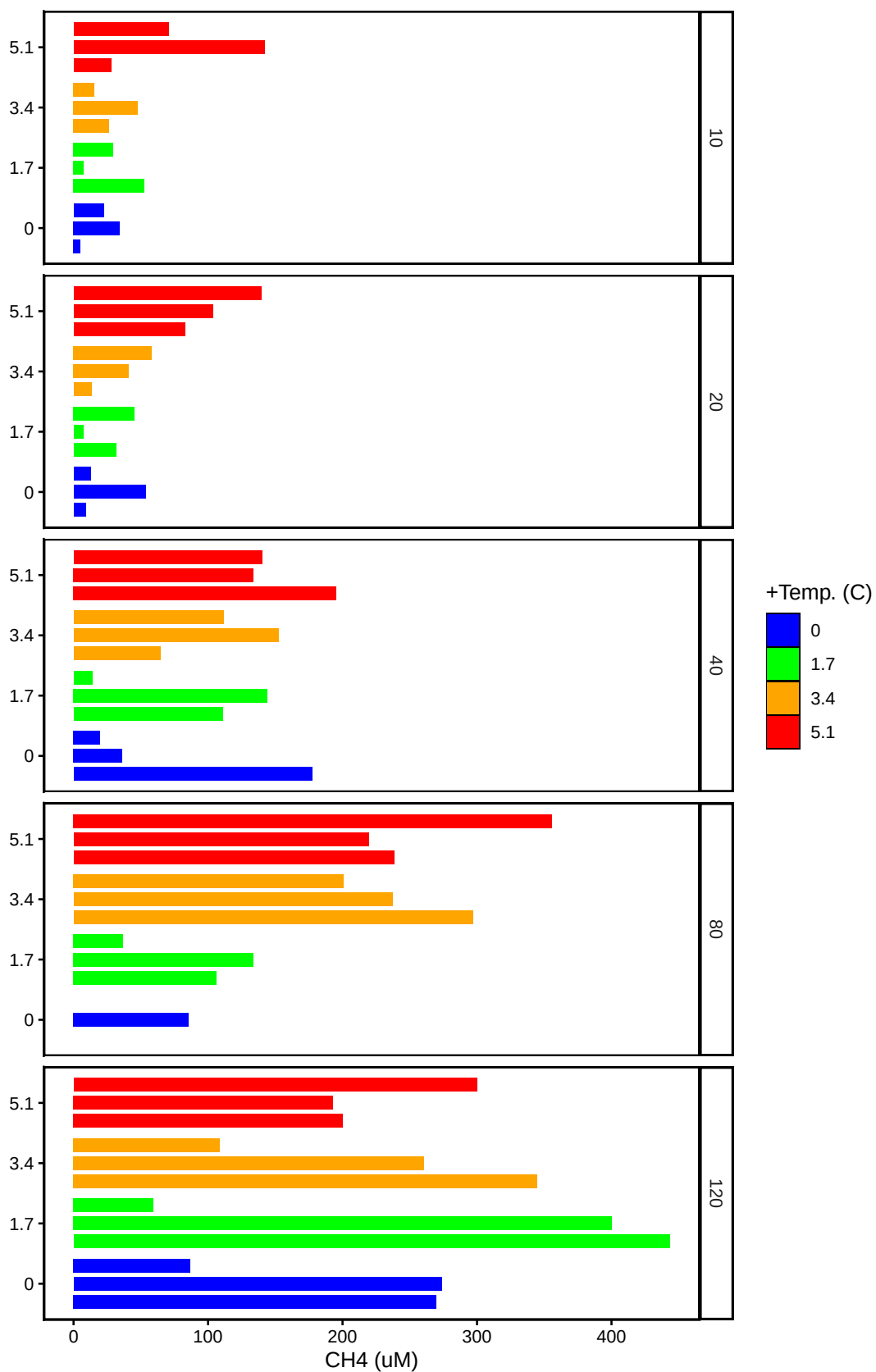
7.2 Visualize Data

```
## Visualize Data
```

C3: Porewater CH4

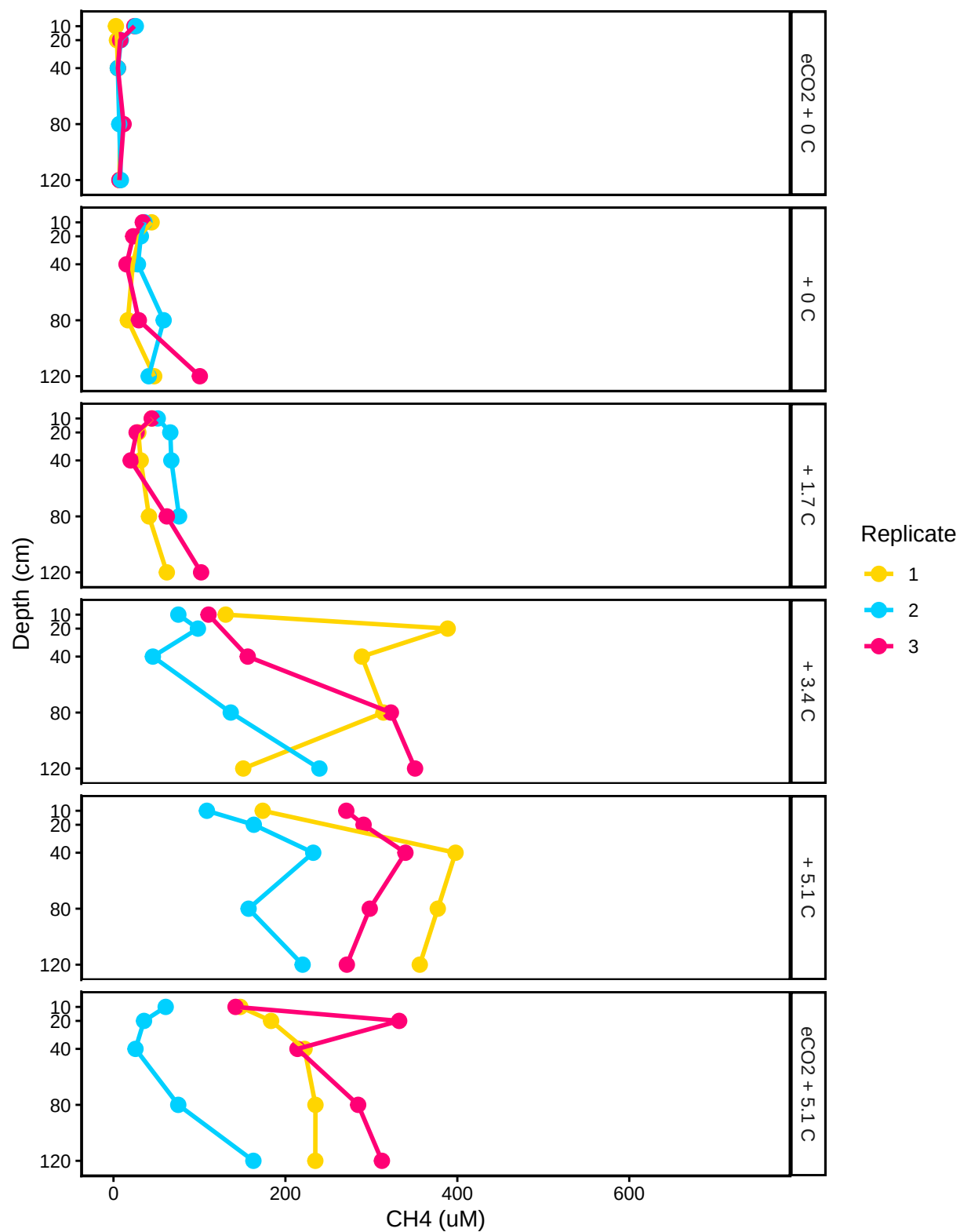


C4: Porewater CH4

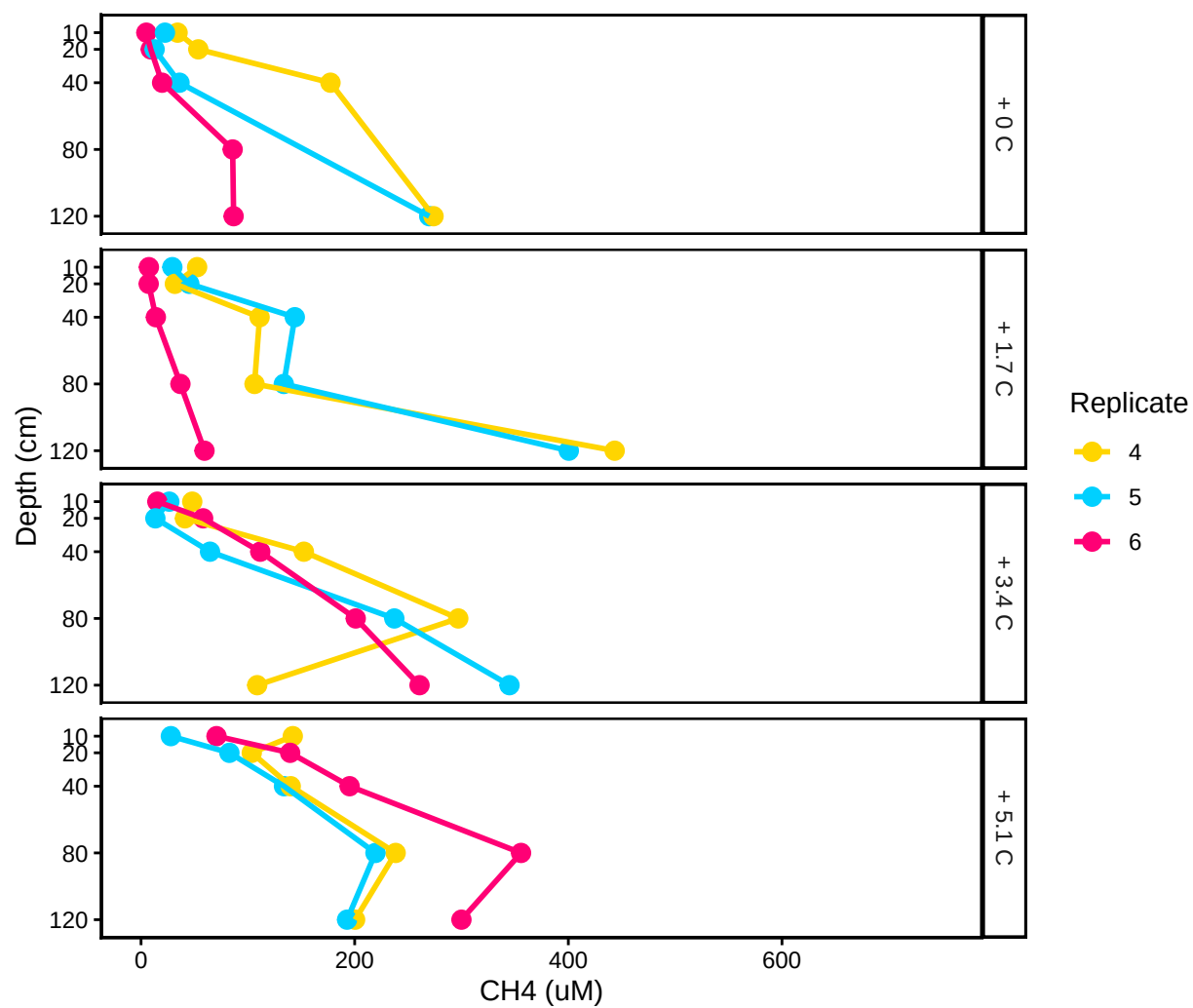


```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.  
## i Please use 'linewidth' instead.  
## This warning is displayed once per session.  
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was  
## generated.
```

C3: Porewater CH₄ by Replicate



C4: Porewater CH4 by Replicate



##Export Data

#end